

Name Class Date

1.

Architecture Quiz

2. Which of the following statements is true?

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|---|---|
| a) The best way to improve a degraded architecture, is to keep fixing the code until the system starts looking and behaving as expected | b) Prescriptive architecture and descriptive architecture are typically the same. |
| c) Architectural drift results in unnecessarily complex architectures. | d) Architectural erosion is less problematic than architectural drift. |

3. Which architectural pattern is best suited for developing a video streaming platform, where various independent services handle distinct functionalities such as content recommendation and user authentication?
 - a) Layered Architecture
 - b) Microkernel
 - c) Microservices
 - d) Serverless
4. What architectural style is most appropriate for an e-commerce platform where multiple events occur simultaneously, such as item search, cart modification, and order placement, and the system must react to these events asynchronously?
 - a) Layered
 - b) Event Driven
 - c) 3-Tier
 - d) Microkernel
5. Which of the following best describes the concept of "cruft" in software architecture?
 - a) Code elements that make it difficult for developers to understand and modify the software.
 - b) Clean and well-organized code that is easy to maintain.
 - c) Temporary files generated by the system during execution.
 - d) Documentation that is outdated and irrelevant to the current system design.
6. Which architecture style would you recommend for building a function-based application like a photo-sharing platform where backend services scale automatically and respond to events such as file uploads?
 - a) Event Driven
 - b) Serverless
 - c) Peer-to-Peer
 - d) Microkernel
7. Which of the following best describes a microkernel architecture?
 - a) An architecture that processes data as a series of transformations.
 - b) A core system that is extended with plug-ins to handle additional functionality.
 - c) A system where all processing happens in a centralized server.
 - d) An architecture where services are split into small, independent modules.

8. A game needs to create different types of obstacles based on the current level, and the exact type of obstacle is not known ahead of time. Which design pattern is most appropriate for handling object creation?
 - a) Observer
 - b) Factory Method
 - c) Singleton
 - d) Decorator
9. A shared document application displays the same document in several views: a text editor, an outline panel, and a word-count panel. Whenever the document changes, all currently open views should automatically update. Views may also be opened or closed while the application is running. Which design pattern best supports this design?
 - a) Observer
 - b) Factory Method
 - c) Decorator
 - d) Strategy
10. A team is designing a food-delivery application. They want one diagram showing the **Order Service, Payment Service, Restaurant Service, and Notification Service and the dependencies among them**. Which UML diagram would be most appropriate?
 - a) Component Diagram
 - b) Deployment Diagram
 - c) Use Case Diagram
 - d) Sequence Diagram
11. A team has finished designing a web application and now wants to document that the frontend runs on a web server, the backend API runs on a cloud application server, and the database runs on a separate database server.
 - a) Component Diagram
 - b) Class Diagram
 - c) Package Diagram
 - d) Deployment Diagram
12. A navigation application supports several ways of calculating a route: fastest route, shortest route, and lowest-cost route. The user can switch between these options while using the application, and adding a new routing algorithm should not require changing the navigation class. Which design pattern is most appropriate?
 - a) Observer
 - b) Decorator
 - c) Factory Method
 - d) Strategy

13. A team is creating a deployment diagram for an online bookstore. The system includes an **AWS application server** and a **BookStoreAPI.jar** file that executes on that server.

How should these two elements be represented?

- a) Both are nodes
- b) The AWS application server is an artifact, and BookStoreAPI.jar is a node.
- c) The AWS application server is a node, and BookStoreAPI.jar is an artifact.
- d) Both are artifacts

Answer Keys

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|--|--|---------------------|
| 1. n/a | 2. c) Architectural drift results in unnecessarily complex architectures. | 3. c) Microservices |
| 4. b) Event Driven | 5. a) Code elements that make it difficult for developers to understand and modify the software. | 6. b) Serverless |
| 7. b) A core system that is extended with plug-ins to handle additional functionality. | 8. b) Factory Method | 9. a) Observer |
| 10. a) Component Diagram | 11. d) Deployment Diagram | 12. d) Strategy |
| 13. c) The AWS application server is a node, and BookStoreAPI.jar is an artifact. | | |

